Jojy John Ph.D.

Clemson University
Post-Doctoral Fellow ,Campbell Lab
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EDUCATION & TRAINING

Postdoc 2023-Present Clemson University, Drivers of functional redundancy across microbiome ; ADvisor: Dr Barbara Campbell

Ph.D 2021 SIST, Chennai, India, Genomic studies and resistance mechanisms of moderate halophilic bacteria ; Advisors: Dr. Amit Kumar and Dr. Vinu Siva

Masters in Data Science 2022 Simplilearn, Bangalore, India

Masters in Science in Microbiology 2014 Mahatma Gandhi University, Kottayam, India

Bachelors in Science in Microbiology 2012 Mahatma Gandhi University, Kottayam, India

Research and Professional Experience

2023 - Present NSF Post doctoral Fellow, Department of Biological Science, Clemson University
2021–2023 Career break (maternity leave), completed a one-year online Master's in Data Science to strengthen computational skills.
2019 -2021- Project Asistant II, Bioinformatics Lab, National Institute of Oceanography, Goa, India
2016 –2019 Seniour anf Junior Research Fellow (Ph.D), SIST, Chennai, India

2014 (summer) Research assistant, RGCB, Tvm, India

2012 (Summer) Research assistant, SSC, Kalady, India

Publications

1. Jojy John and Ramadoss, D.(2025) Genomic insights into microbial resilience to monsoon induced hypoxia inthe Arabian sea:Insights from grnome resolved metagenomic analysis:2025: under revision; World Journal of microbiology and Biotechnology.
2. Ramadoss, D., Ammanabrolu, B. S., Acharya, A., **John, J.**, & Ingole, B. (2025). Deep-sea Life associated with sediments and polymetallic nodules from the Central Indian Ocean Basin: Insights from 18S metabarcoding. Deep Sea Research Part II: Topical Studies in Oceanography, 105487.
3. Ramadoss, Dineshram, Akhil Biju, Chayanika Rathore, Mahua Saha, Prabhu Kolandhasamy, Chandramohan Palogi, **Jojy John,** and Anil Kumar Behera. "The first report on emerged

microplastics in deep-sea sediment: Insights from the Central Indian Ocean Basin." Marine Pollution Bulletin 211 (2025): 117435. (impact factor:5.3)

4. **John, J.,** Dineshram, R., Hemalatha, K. R., Dhassiah, M. P., Gopal, D., & Kumar, A.(2020). Bio-Decolorization of Synthetic Dyes by a Halophilic Bacterium *Salinivibrio* sp. Frontiers in Microbiology, 11, 3281. (Impact factor 5.6)
5. **John, J.,** Siva, V., Kumari, R., Arya, A., & Kumar, A. (2020). Unveiling Cultivable and uncultivable Halophilic Bacteria Inhabiting Marakkanam Saltpan, India and Their Potential for Biotechnological Applications. Geomicrobiology Journal, 1-11. (Impact factor: 2.3)
6. **John, J.,** Siva, V., Richa, K., Arya, A., & Kumar, A. (2019). Life in High Salt Concentrations with Changing Environmental Conditions: Insights from Genomic and Phenotypic Analysis of *Salinivibrio* sp. Microorganisms, 7(11), 577. (Impact factor: 4.152)
7. **John** J, Siva V, Kumar A. (2020). Isolation and characterization of Saltpan bacteria for developing bacterial consortium for the complete removal of nitrogen pollution from aquaculture pond. Conference proceedings – An Insight on Environmental Toxicology (ISBN 978-93-5396-157-2). Hosur, India: MGR College, p377–388
8. **John, J.,** Siva, V. S., & Kumar, A. (2020). Physiological tolerance of the early life history stages of freshwater prawn (*Macrobrachium rosenbergii* De Man, 1879) to environmental stress. Indian Journal of Geo-Marine Sciences Vol. 49 (03), March 2020, pp. 382-389

Preprints and in Progress

1. **Jojy John**, Maximiliano Ortiz, Pierre Ramond, and Barbara J Campbell (2025). "Functional redundancy and metabolic flexibility of microbial communities in two Mid-Atlantic bays" _under review in ISME Communications Journal (round 2)
2. Mir Alvee Ahmed, **Jojy John**, Barbara J. Campbella (2025) Ecological distribution, environmental roles and drivers of Actinobacteriota in two Mid-Atlantic estuaries; under review AEM.
3. **Jojy John,** Dineshram,R (2025) Phototrophy and Chemolithotrophy drive microbial resilience to monsoon hypoxia in the Arabian Sea: Insights from genome-resolved metagenomic analysis _under reviw in Frontier's in Microbiology (Round 1)
4. **Jojy John**†,Manigundan K†, Radhakrishnan M , Amit Kumar, Abirami B, Parli VB (2025) Comparative genomic and phenotypic analysis of *Streptomyces rochei* SOSIST-3 isolated from the Southern Ocean under revision in Polar Biology (Round1)
5. Trisha S, Unnikrishnan K, Satwik M, **Jojy John,** Charles V, Zhixuan F, Marie B, Jennifer R[1], Saji George (2025) Nanoliposome Mediated Co-delivery of Amoxicillin and Tazobactam Remediate Intracellular Infection by a Multidrug-Resistant *Salmonella enterica serovar* Typhimurium under review in Nature Journal of Antibiotics (Round 3, final minor revisions).

Skills

Computational: Proficient in Python, R, and Bash scripting with extensive experience in Linux/Unix environments. Skilled in applying bioinformatics tools and pipelines for 16S amplicon, whole-genome, metagenomic, metatranscriptomic, and virome analyses. Experienced with high-performance computing, job scheduling, scaling, and parallelization of computational workflows. Strong background in multi-omics data integration and large-scale analyses across diverse sample types, ranging from ocean microbiomes to the human gut. Well-versed in handling and analyzing both long-read (Oxford Nanopore) and short-read (Illumina) sequencing data.

Molecular Biology: Experienced in DNA/RNA extraction and purification, Illumina library preparation, quantitative PCR, and microbial cultivation under sterile conditions. Since 2023, I have been working full-time as a bioinformatician, applying these molecular foundations to computational analyses.

Interdiciplinary reaserch collaborations

1. Arctic Microbiome (16S Amplicon) and Actinobacteria & Biosynthetic Potential (India)
2. Radiobacillus Genomics (Chile)
3. Himalayan Microbiome (16S Amplicon_India)
4. Nanoplastic and human microbiome (WGS, 16S amplicon, and shot gun metagenome_ Canada)

Abstract(selected since 2018)

1. Alisha M. Paul, **Jojy John,** Diptee Chaulagain, David Karig, Barbara J. Campbell. 2025 Functional redundancy of marine synthetic biofilm communities under different environmental stresses. ASM biofilm, Oregon, USA
2. **Jojy John**, Mir Alvee Ahmed, Barbara J. Campbell. 2025. Metabolic flexibility, secondary metabolites and seasonal dynamics in particle-associated and free-living microorganisms in the Chesapeake and Delaware Bays, ASM, Los Angles, USA
3. Barbara J. Campbell, Kasey Kiser, Sam Stuckert and **Jojy John.** 2025. Bacteriophages and Auxiliary Metabolic Gene interactions in microbial adaptations in the Chesapeake and Delaware Bays
4. **Jojy John**, Maximiliano Ortiz, and Barbara J Campbell **2025.** Functional Redudancy and metabolic flexibility of microbial communities in two mid Atlantic Bays. ASLO Aquatic Sciences Metting, Charloette, USA
5. Jojy John, Maximiliano Ortiz, Pierre Ramond and Barbara J. Campbell (2024) Microbial functional redundancy in response to substrate and energy utilization in estuarine ecosystems 19th International Symposium on Microbial Ecology (ISME19). · Cape Town, South Africa.
6. Jojy John, Maximiliano Ortiz, Barbara J. Campbell **2024.** Does Microbiome Insure? Clemson University 3rd postdoctoral symposium; Clemson, South Carolina, USA
7. **John, J.,** Siva, V., Kumar A. **2018.** Heavy metal bioremediation potential of moderately halophilic bacteria isolated from salt pan. ICWEE, Sathyabama, 2018. Abstract book: 31. **(Best poster award)**
8. **John, J.,** Siva, V., Kumar A. **2018.** Biotechnological Potential of Moderately Halotolerant Bacteria: Insights from Genomic Analysis of *Salinivibrio* sp. Isolated from Salt Pan. Proceedings of the National Symposium on the Application of Genomics and Proteomics in Aquaculture, Fisheries, & Marine Biology (omicsAFM 2018), Sathyabama Institute of Science and Technology, 13. **(Best oral presentation award)**

Mentoring

2025 Fall 1 Under graduate student (Mary Elizabeth)
2025 Present Bioinformatics support and training to 1 graduate student (Unnikrishnan K) at McGill University, Canada

2025 present Bioinformatics support and training to 1 graduate student (Shilpa Sunny) at SIST, Chennai, India
2023-Present MSc research student in Clemson University (Alisha Paul)
2023- Present 5 graduate students from Clemson University (Dinuka J, Alvee Ahmed, Nichole G, Sulemen M, M Nawas)
2023 –Present Summer Intern Bioinformatics trainer for 1 undergraduate student, Clemson University (Noah)
2019-2021 3 BTech and 4 MTech dissertation intern students, National Institute of Oceanography, Goa, India
2016-2019 2 masters students, SIST, Chennai, India

Teaching

2024-2025 Teacher, Readings in Ecology, Clemson University
2023-2024 Guest Lecturer , Hands on Training on metagenome analysis, MICR 8130: Practical Bioinformatics for Microbiologists, Clemson University, USA
2018-2019 Lab assistant, General Microbiology practical, SIST, Chennai, India
2013-2014 Lab Asistant, Microbiology/biotechnology, Mar Athanasius College, Kothamangalam, India

Honors and Other Awards

Best poster award in International Conference on Waste, Energy, and Environment (ICWEE) 2018
Best Oral presentation in OMICS AFM 2018 (National Symposium on the Application of 2018 Genomics and Proteomics in Aquaculture, Fisheries, and Marine Biology)
6th University rank in Master of Science (Microbiology), MG University, Kerala, India. 2014
10th University rank in Bachelors of Science (Microbiology), MG University, Kerala, India.2012

Public Outreach

2024 **"Ecosystem Resilience of Microbial Communities in the Context of Climate change**", DS College of Arts& Science for women, Tamil Nadu, India
2020 "The magic of eDNA for laypersons";Coastal Impact (NGO), India
2020 **"How Microbial genomics can be applied in Research?"** Center for Bioinformatics, Computational and Systems Biology, Pathfinder Research and Training Foundation, India

PEER REVIEW

-19 Manuscripts from similar reserch background

Microbiology Spectrum,Geomicrobiology Journal,Plos One,Applied and Environmental Microbiology,Scientific Reports,Archives of Microbiology,mSphere,Frontiers in Microbiology, World Journal of microbiology and Biotechnology

PROFESSIONAL REFERENCES

1. **Dr. Barbara Campbell**
   -Relationship: Post doc Advisor

-Position: Dean's Distinguished Professor
-Institution: Clemson University
-Email: *bcampb7@clemson.edu*

-Phone number: **w** 864-656-0559 **c** 302-841-8171

2. **David Karig, Ph.D**
   -Relationship : Advisor_DARPA_post doc
   -Position:Associate Professor
   -Institution: Clemson University
   -Email: *dkarig@clemson.edu*
   -Phone number: **w** 8646565003 **c** 4432807975

3. **Dr. Amit Kumar**
   -Relationship : Advisor Ph.D
   -Position:Scientist
   -Institution: Sathyabama Institute of Science and Technology, Chennai, India-600119
   -Email: amit.kumar.szn@gmail.com/ amitkumar.cccs@sathyabama.ac.in
   -Phone number: +91 9094486533